5(a)(i)	C_2H_3O	**1**

5(a)(ii)	reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W	**4**
	Na	fizz / bubbles etc / **OR** solid / Na disappears / dissolves	$CH_2C(CH_3)CO_2Na$ **OR** $CH_2C(CH_3)CO_2^-Na^+$	
	Br_2 in the dark	brown to colourless / decolourise	$CH_2BrCBr(CH_3)CO_2H$	

Examiner report missing
9701_s25_qp_23 Question 5

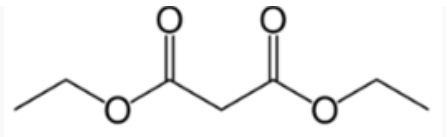
5(a)	$C_3H_4O_3$	**1**

5(b)(i)	$5.00 \times 10^{-3} \times 36.65 / 1000$ $= 1.833 \times 10^{-4} / 1.8325 \times 10^{-4}$	**1**

5(b)(iii) **1**

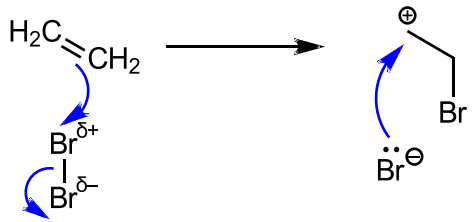
absorption / cm^{-1}	present in spectrum of vitamin C	present in spectrum of **M**
1500–1680	✓	× / – / blank
2850–2950	✓	✓
3200–3650	✓	✓

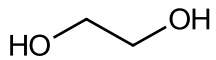
Examiner report missing
9701_s25_qp_22 Question 5

5(a)	**M1** two ester links **M2** rest of structure correct 	**2**

5(d)(i)	**M1** NaOH(aq) **AND** heat **M2** acidified **AND** potassium dichromate(VI) **AND** heat under reflux	**2**

Examiner report missing
9701_s25_qp_21 Question 5

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(c)(i)		**1**
	bromoethene	**1**

4(c)(iii)	C_2H_2	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(b)(i)	step 1 …..Br_2 (in the dark) step 2 …..NaOH(aq) + heat	**2**
5(d)(i)	**W** =	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

Question 5

(a) (i) Those candidates with a good understanding of mass spectrometry realised that the peaks shown in Table 5.1 represented the molecular ion peak, M^+, and its $(M+1)^+$ peak and used the appropriate formula to calculate the relative abundance of the M+1 peak. Some weaker responses based their calculation on an incorrect formula.

(ii) The strongest responses engaged with the information about compound **W**. Weaker responses did not engage fully with the question details and answered in terms of structural formulae. Some formulae gave the fragments with no charges or negative charges.

(b) (i) An understanding that an O–H bond, due to a carboxyl functional group, produces a broad stretch within the range 2500–3000 cm^{-1} was seen in the best responses. The weakest responses did not engage with the details given for compound **W** and its reaction to produce **X**, and listed every potential functional group which might account for bond **A** and bond **B**, including amide and ester functional groups.

(ii) Of those candidates that correctly identified **B** as a C=O of a carbonyl group, some candidates ignored the reaction conditions stated in the question and represented **X** as an aldehyde rather than a ketone.

(c) (i) A significant proportion of candidates correctly named the functional group present in **Y**, based on the information provided.

(ii) Many equations were shown with the correct formula of the organic product; not all equations were balanced. Identification of H_2O rather than H_2 as the other product was seen.

(iii) Those candidates who applied knowledge of the behaviour of different types of alcohol and carbonyl compounds correctly deduced the structure of **Y**. Confusion about the chemical reactions of alcohols and carbonyl compounds was seen in weaker responses.

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	**1**
6(b)(i)	reducing agent for C=O **AND** C=C	**1**
6(b)(ii)	sodium borohydride / $NaBH_4$	**1**
6(c)(i)	**M1** deduce n = 12 (from **Y** – 1C (in methanol)) **M2** $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$	**2**
6(c)(ii)	<table><tr><td></td><td>observation on addition of 2,4–DNPH</td></tr><tr><td>**Y**</td><td>orange precipitate</td></tr><tr><td>**Q**</td><td>no precipitate</td></tr></table> Both correct for one mark	**1**

Question 6

(a) (i) The definition of stereoisomerism was one of the least well answered questions. The most common error was to state that 'molecules had the same molecular formula but different structural formula'. Few mentioned that it was the arrangement of the atoms or groups in space.

(ii) Very few answers gained full credit here. Some candidates identified a C=C double bond and two chiral centres and then incorrectly stated that this gave (2 × 3 =) 6 stereoisomers instead of using the correct expression 2^3 (= 8).

(iii) The molecular formula of **V** was answered quite well. There were no common errors.

(iv) Many candidates correctly identified an alkene and carbonyl group. The most common error was to omit the ester group and substitute it for an alcohol group or a carboxyl/carboxylic acid group.

(b) (i) The question required an identification of the role of reagent **T**. Candidates who stated that **T** behaved as a reducing agent gained full credit. Those candidates who ventured to explain the behaviour of **T** with each functional group generally understood that it reduced the carbonyl group in **V** to an alcohol group in **W**. The reaction of the C=C to from a saturated chain attracted a variety of incorrect answers ranging from acting as a catalyst to hydrolysing the alkene group.

(ii) This question was well answered.

(c) (i) Candidates found this very challenging. Many answers used the m/e values of 200 and 201, from the mass spectrum, instead of the relative abundance values of 100 and x, and attempted to average these values to determine the relative atomic mass.

Creditworthy answers deduced that **Q** contained 12 C atoms and then used this value of *n* in the expression $n = \dfrac{100 \times \text{abundance of M} + 1 \text{ ion}}{1.1 \times \text{abundance of M} + \text{ion}}$

to calculate the abundance of the (M + 1) ion which is equivalent to the value of x.

(ii) This question was answered correctly by many candidates with the most common error being to omit 'precipitate' with the orange colour for the observation with 2,4-DNPH with **Y**.

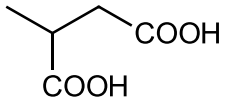
(iii) The majority of candidates did not recognise that the volume of hydrogen gas was collected at s.t.p., in the question, which, from the Table of data, is 22.4 $dm^3 mol^{-1}$.

Various methods were used to calculate the moles of gas produced to include the ideal gas equation and 24.0 $dm^3 mol^{-1}$.

Those candidates who managed to calculate the number of moles of gas then needed to appreciate that the ratio of hydroxyl groups to hydrogen gas was 2 : 1. The award of full credit was rare.

(iv) Many candidates gave correct absorption ranges for one of the two carbonyl absorptions in **Y**, for the ketone and ester functional groups. Several answers gave a range of absorptions for a carboxyl group in **Q** as a difference in the infrared spectrum, having not appreciated that **Q** contains **only** hydroxyl groups, mentioned in the stem of **(c)(ii)**.

2(a)(i)	increases down the group	**1**

2(c)	CH_3OH	**1**
	 one COOH group **AND** correct carbon framework (5C's) (each C has the correct valency)	**1**
	both COOH groups	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

(iii) In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, CO_2, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

(iv) The addition of ethanoic acid to $Ba(OH)_2$ gave barium ethanoate, $(CH_3COO)_2Ba$, as the product.

Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

(v) $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

(b) (i) The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

(ii) The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

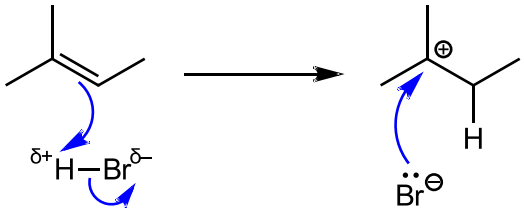
$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

4(a)(i)	3-methylbut-1-ene	**1**

4(c)(i)	 correct dipole on H-Br AND curly arrow from C=C bond to $H^{\sigma+}$ **AND** curly arrow from H—Br bond to $Br^{\sigma-}$	**1**
	correct (Markovnikoff) intermediate	**1**
	curly arrow from lone pair on Br^- to C^+ of intermediate	**1**

4(d)	(reaction 2) NaOH**(aq)**	**1**
	(reaction 3) hot **acidified <u>concentrated</u>** $KMnO_4$	**1**
	(reaction 4) <u>conc</u> H_2SO_4 catalyst	**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

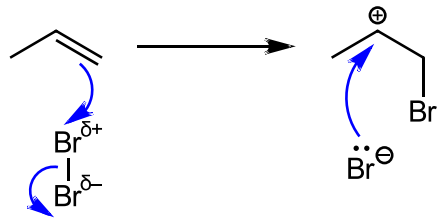
The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified <u>concentrated</u> $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed <u>concentrated</u> H_2SO_4 (as a catalyst).

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(b)(i)	 curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond	**1**
	correct intermediate	**1**
	curly arrow from lone pair on Br^- to $C^{(+)}$ of the intermediate	**1**

4(b)(iii)	HCN&KCN **OR** NaCN with H_2SO_4	**1**
	NaOH(aq) (1) dilute / aq(ueous)HC*l* / H_2SO_4 (1)	**2**
	acidified $K_2Cr_2O_7$	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

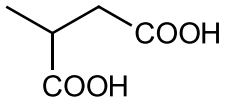
(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

2(a)(i)	increases down the group	**1**

2(c)	CH_3OH	**1**
	 one COOH group **AND** correct carbon framework (5C's) (each C has the correct valency)	**1**
	both COOH groups	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

(iii) In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, CO_2, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

(iv) The addition of ethanoic acid to $Ba(OH)_2$ gave barium ethanoate, $(CH_3COO)_2Ba$, as the product.

Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

(v) $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

(b) (i) The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

(ii) The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(i)	1 4 0	**1**

3(e)(ii)	Solution turns from purple to colourless	**1**
	Orange / red precipitate forms	**1**
	no reaction	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

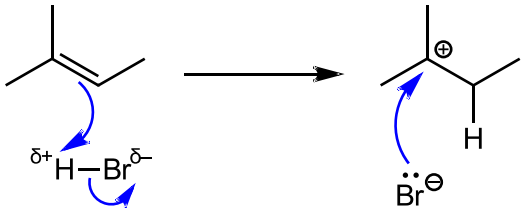
Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

4(a)(i)	3-methylbut-1-ene	**1**

4(c)(i)	 correct dipole on H-Br AND curly arrow from C=C bond to $H^{\sigma+}$ **AND** curly arrow from H—Br bond to $Br^{\sigma-}$	**1**
	correct (Markovnikoff) intermediate	**1**
	curly arrow from lone pair on Br^- to C^+ of intermediate	**1**

4(d)	(reaction 2) NaOH**(aq)**	**1**
	(reaction 3) hot **acidified <u>concentrated</u>** $KMnO_4$	**1**
	(reaction 4) <u>conc</u> H_2SO_4 catalyst	**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified concentrated $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed concentrated H_2SO_4 (as a catalyst).

5(a)	C_3H_7O	**1**

5(d)				**3**

reagent	observation on addition to **X**	observation on addition to **Y**
sodium carbonate solution	effervescence	no change
2,4-DNPH	orange precipitate	no change
alkaline aqueous iodine	yellow precipitate	yellow precipitate

Two correct for one mark, four correct for two marks, six correct for three marks

Question 5

(a) Some candidates deduced the correct empirical formula of C_3H_7O from a molecular formula of $C_6H_{14}O_2$.

(b) (i) Very few candidates used molecular formula, as required in the question, to construct an equation for the reduction reaction.

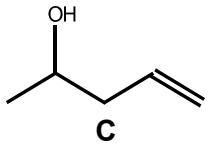
(ii) Many answers correctly stated that a suitable reducing agent for the reaction is $LiAlH_4$.

(c) Some good answers were seen.

(d) Many candidates correctly stated that **X**, the ketoacid, would react with all three reagents to give effervescence, an orange precipitate and yellow precipitate respectively. Fewer candidates recognised that the diol, **Y**, would only react with the alkaline $I_2(aq)$ as it contained a $CH_3CH(OH)$ moiety.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**

5(b)(i)	carbonyl (group)	**1**

5(c)(i)	 **C** **M1** $CH_3CH(OH)$ present **M2** five carbons in a straight chain **AND** molecular formula $C_5H_{10}O$ **AND** 1 C=C structure which shows only one type of stereoisomerism	**2**

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.